

RESEARCH INTERESTS

- Scientific Machine Learning
- Learning-to-Optimize
- Model Predictive Control
- Cyber-Physical-Human Systems
- Robotics and Autonomous Systems
- Distributed Optimization and Multi-Agent Systems

EDUCATION

University of Delaware Ph.D. in Mechanical Engineering	Newark, DE USA Aug. 2021–Aug. 2025
Northern Arizona University M.Sc. in Informatics (with Distinction)	Flagstaff, AZ USA Aug. 2019–May 2021
Hanoi University of Science and Technology B.Sc. in Control Engineering and Automation	Hanoi, Vietnam Aug. 2014–Jun. 2019

RESEARCH EXPERIENCE

University of Pennsylvania Postdoctoral Researcher, Department of Electrical and Systems Engineering – Learning-to-Optimize and Its Applications in Control	Philadelphia, PA USA Aug. 2025–Present
Cornell University Visiting Graduate Scholar, Systems Engineering – Learning and Control for Connected and Automated Vehicles in Mixed Traffic – Development of IDS’s Scaled Smart City Robotic Testbed (Version 2.0)	Ithaca, NY USA Aug. 2023–Aug. 2025
University of Delaware Graduate Research Assistant, Department of Mechanical Engineering – Learning and Control for Connected and Automated Vehicles in Mixed Traffic	Newark, DE USA Aug. 2021–May 2023
Northern Arizona University Graduate Research Assistant, School of Informatics, Computing, and Cyber Systems – Learning-Based Model Predictive Control with Gaussian Processes – Adaptive Sampling for Mobile Robotic Sensor Networks	Flagstaff, AZ USA Aug. 2019–May 2021
Vietnam Maritime University Undergraduate Research Intern, School of Mechanical Engineering – Research, Design, and Manufacture of a Floating Crane Testbed	Haiphong, Vietnam Sep. 2017–Mar. 2019

TEACHING EXPERIENCE

University of Delaware Graduate Teaching Assistant, Department of Mechanical Engineering – Vibration and Control (MEEG 312) , Fall 2021 – Laboratory Instructor – Dynamics (MEEG 211) , Spring 2022 – Recitation Instructor	Newark, DE USA Aug. 2021–May 2022
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INDUSTRY RESEARCH EXPERIENCE

Honda Research Institute USA
Graduate Student Intern

Ann Arbor, MI USA
Jun. 2023–Aug. 2023

– Social Navigation for Multiple Robots in Crowded Environments

PUBLICATIONS

JOURNAL ARTICLES

- [1] **V.-A. Le** and A. A. Malikopoulos, “Controller Adaptation via Learning Solutions of Contextual Bayesian Optimization”, *IEEE Robotics and Automation Letters*, vol. 10, no. 8, pp. 8308–8315, 2025.
- [2] **V.-A. Le**, B. Chalaki, F. N. Tzortzoglou, and A. A. Malikopoulos, “Stochastic Time-Optimal Trajectory Planning for Connected and Automated Vehicles in Mixed-Traffic Merging Scenarios”, *IEEE Transactions on Control Systems Technology*, vol. 33, no. 4, pp. 1403–1417, 2025.
- [3] **V.-A. Le** and A. A. Malikopoulos, “Distributed Optimization for Traffic Light Control and Connected Automated Vehicle Coordination in Mixed-Traffic Intersections”, *IEEE Control Systems Letters*, vol. 8, pp. 2721–2726, 2024.
- [4] A. Mokhtarian, J. Xu, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, **V.-A. Le**, S. Ulhas, J. Betz, S. Wilson, *et al.*, “A survey on small-scale testbeds for connected and automated vehicles and robot swarms: A guide for creating a new testbed [Survey]”, *IEEE Robotics & Automation Magazine*, vol. 32, no. 3, pp. 146–163, 2024.
- [5] A. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, “A safety-prioritized receding horizon control framework for platoon formation in a mixed traffic environment”, *Automatica*, vol. 155, p. 111115, 2023.
- [6] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “Multistep Predictions for Adaptive Sampling in Mobile Robotic Sensor Networks Using Proximal ADMM”, *IEEE Access*, vol. 10, pp. 64850–64861, 2022.
- [7] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, “ADMM-Based Adaptive Sampling Strategy for Nonholonomic Mobile Robotic Sensor Networks”, *IEEE Sensors Journal*, vol. 21, no. 13, pp. 15369–15378, 2021.
- [8] **V.-A. Le**, X. H. Le, L. Nguyen, and X. M. Phan, “An efficient adaptive hierarchical sliding mode control strategy using neural networks for 3D overhead cranes”, *International Journal of Automation and Computing*, vol. 16, no. 5, pp. 614–627, 2019.
- [9] H. X. Le, T. V. Nguyen, **V.-A. Le**, T. A. Phan, N. H. Nguyen, and M. X. Phan, “Adaptive hierarchical sliding mode control using neural network for uncertain 2D overhead crane”, *International Journal of Dynamics and Control*, vol. 7, pp. 996–1004, 2019.
- [10] A. T. Le, M. C. Hoang, V. T. Pham, C. N. Luong, D. T. Vu, and **V.-A. Le**, “Adaptive neural network sliding mode control of shipboard container cranes considering actuator backlash”, *Mechanical Systems and Signal Processing*, vol. 112, pp. 233–250, 2018.

CONFERENCE PAPERS

- [1] **V.-A. Le**, R. Tumu, and R. Mangharam, “Toward Single-Step MPPI via Differentiable Predictive Control”, in *2026 IEEE 65th Conference on Decision and Control (CDC)*, (accepted, arXiv preprint arXiv:2604.01539).
- [2] **V.-A. Le** and A. A. Malikopoulos, “Accelerating Time-Optimal Trajectory Planning for Connected and Automated Vehicles with Graph Neural Networks”, in *23rd IFAC World Congress*, (accepted, arXiv preprint arXiv:2511.20383).
- [3] **V.-A. Le**, M. Xie, and R. Mangharam, “A Hybrid Learning-to-Optimize Framework for Mixed-Integer Quadratic Programming”, in *Proceedings of the 8th Annual Learning for Dynamics and Control Conference*, ser. Proceedings of Machine Learning Research, vol. 331, PMLR, 2026, pp. 1055–1069.
- [4] A. Amine, K. Puri, **V.-A. Le**, and R. Mangharam, “Nonplanar Model Predictive Control for Autonomous Vehicles with Recursive Sparse Gaussian Process Dynamics”, in *2026 IEEE Intelligent Vehicles Symposium (IV)*, IEEE, 2026, pp. 2088–2093.
- [5] **V.-A. Le**, P. Kounatidis, and A. A. Malikopoulos, “Combining Graph Attention Networks and Distributed Optimization for Multi-Robot Mixed-Integer Convex Programming”, in *2025 IEEE 64th Conference on Decision and Control (CDC)*, IEEE, 2025, pp. 4440–4445.

- [6] **V.-A. Le**, B. Chalaki, V. Tadiparthi, H. N. Mahjoub, J. D'Sa, and E. Moradi-Pari, "Social Navigation in Crowded Environments with Model Predictive Control and Deep Learning-Based Human Trajectory Prediction", in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 4793–4799.
- [7] **V.-A. Le**, V. Tadiparthi, B. Chalaki, H. N. Mahjoub, J. D'sa, E. Moradi-Pari, and A. A. Malikopoulos, "Multi-Robot Cooperative Navigation in Crowds: A Game-Theoretic Learning-Based Model Predictive Control Approach", in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 4834–4840.
- [8] **V.-A. Le**, H. M. Wang, G. Orosz, and A. A. Malikopoulos, "Coordination for Connected Automated Vehicles at Merging Roadways in Mixed Traffic Environment", in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 4150–4155.
- [9] N. Venkatesh, **V.-A. Le**, A. Dave, and A. A. Malikopoulos, "Connected and Automated Vehicles in Mixed-Traffic: Learning Human Driver Behavior for Effective On-Ramp Merging", in *2023 62nd IEEE Conference on Decision and Control (CDC)*, 2023, pp. 92–97.
- [10] **V.-A. Le** and A. A. Malikopoulos, "Optimal Weight Adaptation of Model Predictive Control for Connected and Automated Vehicles in Mixed Traffic with Bayesian Optimization", in *2023 American Control Conference (ACC)*, 2023, pp. 1183–1188.
- [11] **V.-A. Le** and A. A. Malikopoulos, "A Cooperative Optimal Control Framework for Connected and Automated Vehicles in Mixed Traffic Using Social Value Orientation", in *2022 61st IEEE Conference on Decision and Control (CDC)*, 2022, pp. 6272–6277.
- [12] A. M. I. Mahbub, **V.-A. Le**, and A. A. Malikopoulos, "Safety-Aware and Data-Driven Predictive Control for Connected Automated Vehicles at a Mixed Traffic Signalized Intersection", in *10th IFAC International Symposium on Advances in Automotive Control*, IFAC, 2022, pp. 51–56.
- [13] **V.-A. Le** and T. X. Nghiem, "Distributed Experiment Design and Control for Multi-agent Systems with Gaussian Processes", in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 2226–2231.
- [14] **V.-A. Le** and T. X. Nghiem, "A Receding Horizon Approach for Simultaneous Active Learning and Control using Gaussian Processes", in *2021 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2021, pp. 453–458.
- [15] **V.-A. Le**, L. Nguyen, and T. X. Nghiem, "An Efficient Adaptive Sampling Approach for Mobile Robotic Sensor Networks using Proximal ADMM", in *2021 American Control Conference (ACC)*, IEEE, 2021, pp. 1101–1106.
- [16] **V.-A. Le** and T. X. Nghiem, "Gaussian Process Based Distributed Model Predictive Control for Multi-agent Systems using Sequential Convex Programming and ADMM", in *2020 IEEE Conference on Control Technology and Applications (CCTA)*, IEEE, 2020, pp. 31–36.
- [17] T. X. Nghiem, T.-D. Nguyen, and **V.-A. Le**, "Fast Gaussian Process based Model Predictive Control with Uncertainty Propagation", in *2019 57th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, IEEE, 2019, pp. 1052–1059.
- [18] **V.-A. Le**, X. H. Le, D. T. Vu, V. T. Pham, A. T. Le, and M. C. Hoang, "Designing an adaptive controller for 3D overhead cranes using hierarchical sliding mode and neural network", in *2018 International Conference on System Science and Engineering (ICSSE)*, IEEE, 2018, pp. 1–6.

PATENTS

- [1] **V.-A. Le**, V. Tadiparthi, B. Chalaki, H. N. Mahjoub, J. J. D'Sa, and E. Moradi-Pari, "Systems and Methods for Robot Navigation", U.S. Patent Application Publication US 2025/0117025 A1, U.S. Patent Application No. 18/907,188, Apr. 10, 2025.

HONORS AND AWARDS

- Jun. 2026: **IEEE ITSS Best Dissertation Award (2nd Place)**, IEEE Intelligent Transportation Systems Society
- Feb. 2026: **Penn AI Fellowship**, University of Pennsylvania
- Jul. 2025: **Best Paper Finalist (2nd Place)**, ASME Automotive and Transportation Systems Technical Committee (ATS-TC), 2025 American Control Conference
- Aug. 2023: Student Travel Award, IEEE Control Systems Society, 2023 IEEE Conference on Decision and Control

- Oct. 2022: IEEEExtreme Programming Competition, Global Rank: 949/6373; U.S. Rank: 12/99
- Apr. 2022: Student Travel Awards, University of Delaware Graduate College and IEEE Control Systems Society, 2022 American Control Conference
- Sep. 2021: Student Travel Award, IEEE Control Systems Society, 2021 IEEE Conference on Decision and Control
- Jun. 2021: Student Travel Award, IEEE Control Systems Society, 2021 IEEE Conference on Control Technology and Applications
- Aug. 2020: Student Travel Award, IEEE Control Systems Society, 2020 IEEE Conference on Control Technology and Applications
- Aug. 2019: Presidential Fellowship, Northern Arizona University
- Aug. 2018: Odon Vallet Scholarship for Undergraduate Students
- Jun. 2018: Conference Travel Award, Vietnam National Foundation for Science and Technology Development (NAFOS-TED), 2018 IEEE International Conference on System Science and Engineering
- Apr. 2015: Gold Medal, Vietnam National Mathematical Olympiad for Undergraduate Students

STUDENT RESEARCH MENTORING

- Mu Xie (MS Robotics, University of Pennsylvania) – Learning-to-optimize for mixed-integer optimization
- Kabir Puri (MS Robotics, University of Pennsylvania) – Learning-based control for autonomous driving on nonplanar terrains
- Yuntong Li (MS Electrical Engineering, University of Pennsylvania) – Mixed-integer model predictive control for bipedal locomotion
- Shamanth Sreekanth (MS Electrical Engineering, University of Pennsylvania) – Physics-informed modeling and optimization for process control
- Nandagopal Vidhu (MS Robotics, University of Pennsylvania) – Integration of differentiable robotic simulators into learning-to-optimize frameworks
- Owen Sorber (MS Computer Science, Cornell University) – Development of a containerized Docker-based software stack for advanced driver-assistance systems on autonomous robotic vehicles
- Taylor Stephens (MS Systems Engineering, Cornell University) – Deployment of the Crazyswarm multi-drone platform and development of aerial-ground autonomy demonstrations using Crazyswarm and IDS3C
- Qiming Yang (MS Systems Engineering, Cornell University) – Design and implementation of a digital simulation environment for the IDS3C testbed
- Evans McCleary (BS Mechanical Engineering, University of Delaware) – Development and maintenance of hardware and software infrastructure for robotic vehicles in the IDS3C testbed

INVITED TALKS AND TUTORIALS

1. **Tutorial Co-Organizer and Speaker**, “Learning-to-Optimize for Control in Robotics,” IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), October 2026.
2. **Lightning Talk**, “Learning-to-Optimize and Its Applications in Model Predictive Control,” 3rd Northeast Systems and Control Workshop, May 2026.
3. **Invited Seminar**, “Towards Real-Time Cooperative Trajectory Planning for Connected and Automated Vehicles in Mixed Traffic,” RGSO-CARS Seminar, University of Delaware, May 2025.

PROFESSIONAL SERVICE

- Professional Memberships
 - Member, Institute of Electrical and Electronics Engineers (IEEE) 2020–Present
 - Member, IEEE Control Systems Society 2020–Present

- Member, IEEE Intelligent Transportation Systems Society 2022–Present
- Member, IEEE Robotics and Automation Society 2022–Present
- Technical Committee Memberships
 - Member, IEEE Control Systems Society Technical Committee on Smart Cities 2022–Present
 - Member, IEEE Control Systems Society Technical Committee on Automotive Controls 2024–Present
 - Member, IEEE Robotics and Automation Society Technical Committee on Optimization for Robotics 2026–Present
- Peer Review
 - **Journals:** Automatica; IEEE Transactions on Automatic Control; IEEE Transactions on Robotics; IEEE/ASME Transactions on Mechatronics; IEEE Transactions on Intelligent Transportation Systems; IEEE Transactions on Control Systems Technology; IEEE Transactions on Vehicular Technology; IEEE Transactions on Intelligent Vehicles; IEEE Control Systems Letters; IEEE Robotics and Automation Letters; IEEE Open Journal of Control Systems; Transportation Research Part C: Emerging Technologies; Journal of the Franklin Institute.
 - **Conferences:** IEEE Conference on Decision and Control (CDC); American Control Conference (ACC); European Control Conference (ECC); IEEE International Conference on Robotics and Automation (ICRA); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE International Conference on Intelligent Transportation Systems (ITSC); IEEE Intelligent Vehicles Symposium (IV); Modeling, Estimation and Control Conference (MECC); IFAC World Congress.